

Chapter 2: HTML Basics

What is HTML?

The set of markup symbols or codes placed in a file intended for display on a Web browser page. The World Wide Web Consortium (<http://w3c.org>) sets the standards for HTML and its related languages.

HTML Elements

- Each markup code represents an HTML **element**.
- Each element has a purpose.
Most elements are coded as a pair of tags: an opening tag and a closing tag.
- Tags are enclosed in angle brackets, “<” and “>” symbols.

What is HTML5?

- Newest version of HTML/XHTML
- Supported by modern browsers
- Intended to be backwards compatible
- Adds new elements
- Adds new functionality
- Edit form data
 - Native video and audio
 - And more!
 - Source: W3C <http://www.w3.org/html/logo/>

Document Type Definition

Document Type Definition (DTD)

- doctype statement
- identifies the version of HTML contained in your document.
- placed at the top of a web page document

<!DOCTYPE html>

Example HTML5 Web Page

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Page Title Goes Here</title>
<meta charset="utf-8">
</head>
<body>
... body text and more HTML5 tags go here ...
</body>
</html>
```

- This outline (shell) is given to you in the Class Resources → Web page template (template.html)

Head & Body Sections

Head Section

Contains information that describes the web page document

<head>

...head section info goes here

</head> Body Section

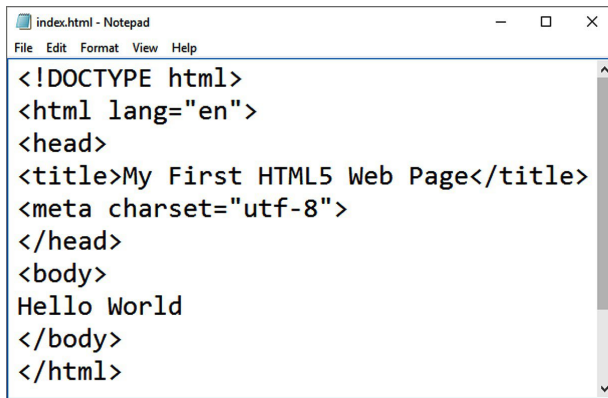
Contains text and elements that display in the web page document

<body>

...body section info goes here

</body>

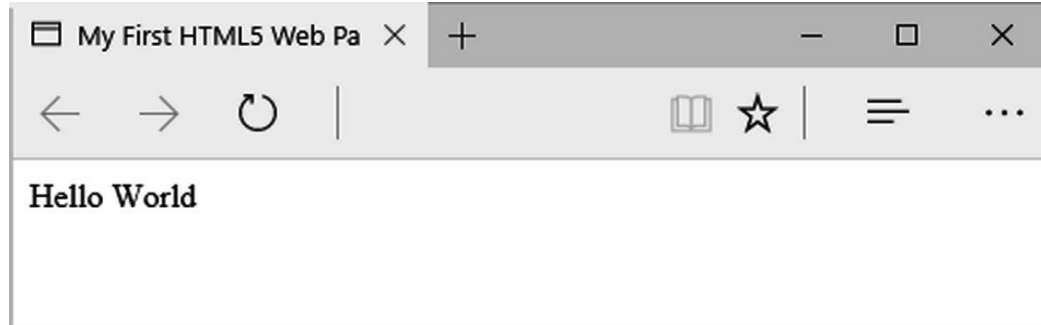
Code displayed in Notepad. Courtesy of Microsoft Corporation.



```
index.html - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html lang="en">
<head>
<title>My First HTML5 Web Page</title>
<meta charset="utf-8">
</head>
<body>
Hello World
</body>
</html>
```

Title Element
Meta Element

Web page displayed by Microsoft Edge. Courtesy of Microsoft Corporation.



Heading Element

`<h1>Heading Level 1</h1>`

`<h2>Heading Level 2</h2>`

`<h3>Heading Level 3</h3>`

`<h4>Heading Level 4</h4>`

`<h5>Heading Level 5</h5>`

`<h6>Heading Level 6</h6>`

Heading Level 1

Heading Level 2

Heading Level 3

Heading Level 4

Heading Level 5

Heading Level 6

Paragraph Element

Paragraph element

`<p> ...paragraph goes here... </p>`

- Groups sentences and sections of text together.
- Block Display – Configures empty space above and below

Line Break Element

Line Break element

- Stand-alone, or void tag
*...text goes here
*
This starts on a new line....
- Causes the next element or text to display on a new line

Blockquote Element

Blockquote element

- Indents a block of text for special emphasis

<blockquote>

...text goes here...

</blockquote>

- Block Display – Configures empty space above and below

Phrase Elements

Indicate the context and meaning of the text

Element	Example	Usage
<code></code>	bold text	Text that has no extra importance but is styled in bold font by usage and convention
<code></code>	emphasized text	Causes text to be emphasized in relation to other text; usually displayed in italics
<code><i></code>	italicized text	Text that has no extra importance but is styled in italics by usage and convention
<code><mark></code>	mark text	Text that is highlighted in order to be easily referenced (HTML5 only)
<code><small></code>	small text	Legal disclaimers and notices ("fine print") displayed in small font-size
<code></code>	strong text	Strong importance; causes text to stand out from surrounding text; usually displayed in bold
<code><sub></code>	_{sub} text	Displays a subscript as small text below the baseline
<code><sup></code>	^{sup} text	Displays a superscript as small text above the baseline

Proper Nesting

CODE:

```
<p><i>Call for a free quote for your web development needs: <strong>888.555.5555 </strong></i></p>
```

BROWSER DISPLAY:

Call for a free quote for your web development needs: 888.555.5555

HTML Lists

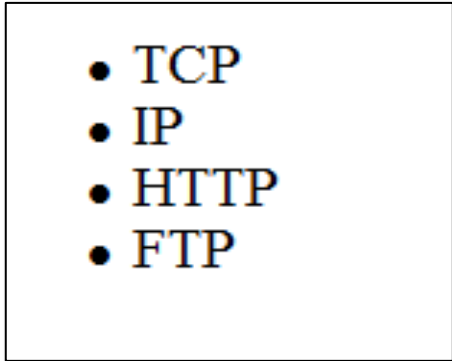
Unordered List Ordered List Description List

Unordered List

Displays a bullet, or list marker, before each entry in the list.

Contains the unordered list

Contains an item in the list

- 
- TCP
 - IP
 - HTTP
 - FTP

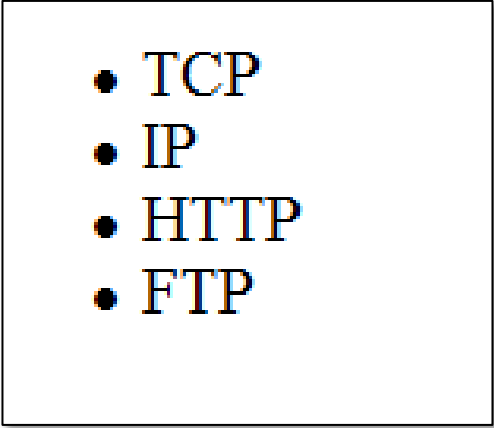
Unordered List Example

TCP

IP

HTTP

FTP

- 
- TCP
 - IP
 - HTTP
 - FTP

Ordered List

Displays a numbering or lettering system to itemize the information contained in the list

Contains the ordered list

- type attribute determines numbering scheme of list, default is numerals

Contains an item in the list

Ordered List Example

Apply to school

Register for course

Pay tuition

Attend course

1. Apply to school
2. Register for course
3. Pay tuition
4. Attend course

Description List

Useful to display a list of terms and descriptions or a list of FAQ and answers

- <dl>
Contains the description list
- <dt>
Contains a term/phrase/sentence
Configures empty space above and below the text
- <dd>
Contains a description of the term/phrase/sentence
 - Indents the text
 - Configures empty space above and below the text

Description List Example

<dl>

<dt>IP</dt>

<dd>Internet Protocol</dd>

<dt>TCP</dt>

<dd>Transmission Control Protocol</dd>

</dl>

IP

Internet Protocol

TCP

Transmission Control Protocol

Special Characters

- Display special characters such as quotes, copyright symbol, etc.

Character

©

<

>

&

Code

©

<

>

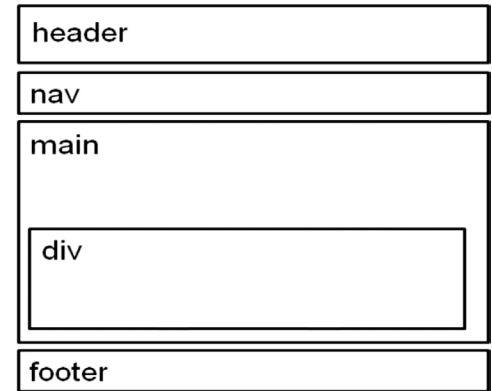
&

Div Element

Configures a structural block area or “division” on a web page with empty space above and below.

Can contain other block display elements, including other div elements

```
<div>Home Services Contact</div>
```



HTML5 Structural Elements (1 of 3)

header Element

```
<header></header>
```

Contains the web page document's headings

nav Element

```
<nav></nav>
```

Contains web page
document's main navigation

HTML5 Structural Elements (2 of 3)

main Element

```
<main></main>
```

Contains the web page document's main content

footer Element

```
<footer></footer>
```

Contains the web page document's footer

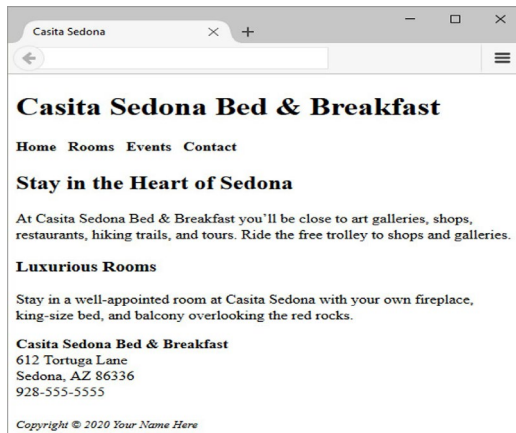


Figure 2.21 Casita Sedona web page

HTML5 Structural Elements (3 of 3)

<body>

<header> *document headings go here* </header>

<nav> *main navigation goes here* </nav>

<main> *main content goes here* </main>

<footer> *document footer information goes here*
</footer>

</body>

A Element (Anchor Element)

- Specifies a hyperlink reference (href) to a file
- Text between the <a> and is displayed on the web page.

Contact Us

- href Attribute
 - Indicates the file name or URL

Opening a Link in a New Browser Window

The target attribute on the anchor element opens a link in a new browser window or new browser tab.

Search Google

HTML5 Block Anchor

Configure block display elements within a hyperlink

`<a href="http://www.w3.org/TR/html-markup">`

`<h1>HTML5 Reference</h1>`

`<p>Bookmark this site for a handy HTML5 reference.</p>`

`</a>`



Absolute & Relative Hyperlinks

Absolute link

- Link to a different website

`<a href="http://yahoo.com">Yahoo</a>`

Relative link

- Link to pages on your own site

`<a href="index.htm">Home</a>`

E-Mail Hyperlink

Automatically launch the default mail program configured for the browser

If no browser default is configured, a message is displayed

```
<a ref="mailto:me@gmail.com">me@gmail.com</a>
```